

FGI's GNSS Jamming and Spoofing Dataset Repository (FGI-JSDR)

Welcome to FGI's GNSS Jamming and Spoofing Dataset Repository (FGI-JSDR) – a unique resource for advancing the research and development toward resilience against jamming and spoofing. In today's landscape, the vulnerability of GNSS receivers and GNSS-dependent systems to jamming or spoofing attacks is on the rise. To address this, we present a collection of open datasets featuring raw GNSS In-phase and Quadrature (I/Q) data.

Datasets:

1. *FGI's GNSS Spoofing Dataset Repository (FGI-SpoofRepo) can be accessed through the following link: **FGI-SpoofRepo** (<https://etsin.fairdata.fi/dataset/367379a8-7d78-4b08-91f0-8027ce7a621b>).*

*Enriched with today's real-world modernized GNSS signals and cryptographic signatures (with Galileo E1B signal), these datasets offer an ideal foundation for assessing the performance of GNSS receivers under different malicious environment with the presence of jamming and/or spoofing. Click on the link above to access these datasets for your research and development projects. These datasets include signals from GPS L1 C/A, Galileo E1, GPS L5, and Galileo E5a signals, covering three distinct types of spoofing characteristics: synchronous, asynchronous, and meaconing. **Here is the accompanying paper in GPS Solution** (<https://link.springer.com/article/10.1007/s10291-024-01719-2>).*

2. *FGI OSNMA datasets can be accessed through the following link: **FGI OSNMA** (<https://etsin.fairdata.fi/dataset/09dc5c1b-933d-4efd-aa66-be2c07fab3b3>).*

This repository includes raw GNSS In-phase and Quadrature (I/Q) data featuring GPS L1 C/A and Galileo E1 signals. This dataset covers two distinctive scenarios to explore Galileo's Open Service Navigation Message Authentication (OSNMA) feature. Scenario 1 consists of clean GNSS data captured under open sky conditions, while Scenario 2 comprises data with spoofing impairments gathered during the Jammertest 2023 conducted in Andøya, Norway. More information about the dataset can be accessed through the provided link.

3. **FGI Jammertest 2023** datasets can be accessed through the following link: **FGI Jammertest 2023 Repository**. (<https://etsin.fairdata.fi/dataset/2eee8cfd-db00-4e46-bd62-d7632c09c700>).

This repository contains a collection of raw In-phase and Quadrature (I/Q) GNSS data samples that were recorded during a jammer test campaign named 'Jammertest 2023'. This test campaign was hosted in Andøya, Norway. The digitized I/Q data samples were captured via a record and replay device called 'LabSat 3 wideband'. It comprises of live recordings of GPS L1 C/A, Galileo E1 signals. These datasets contain one high power jamming and two other distinct spoofing scenarios (i.e., incoherent and coherent), making them valuable examples for testing the effectiveness of various anomaly detection/mitigation techniques in real world scenarios. More information about the dataset can be accessed through the provided link.

Analyzing Tool:

The open-source software-defined receiver FGI-GSRx can be used to analyze the above datasets. The relevant configuration files to run FGI-GSRx (v2.0.0) are also shared for the users. It is imperative for users to exercise due diligence when utilizing the software-receiver, recognizing potential risks, and modifying them accordingly to align with the specifics of their systems and intended use cases.

- **FGI-GSRx Software Receiver**. (<https://github.com/nlsfi/FGI-GSRx>).

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